SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Synopsis of Major Project-A On

Threat'Sentry – Detect, Simulate, and Defend

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CERTIFICATE

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Abstract

As machine learning (ML) systems drive increasingly important applications—self-driving cars to cybersecurity—their susceptibility to adversarial attacks has become a fundamental security issue. Threat'Sentry is an adaptive defense system that detects, mimics, and protects against adversarial manipulations of ML models. The system continuously monitors model inputs and activities to detect fine-grained perturbations that could cause incorrect predictions, thereby maintaining the integrity and reliability of deployed AI systems.

At its heart, Threat'Sentry incorporates a simulation engine that creates and examines adversarial examples for various attack types like Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and Carlini–Wagner (C&W) attacks. By simulating these attacks in test environments, it makes model vulnerabilities understandable to researchers and practitioners, allows them to measure robustness, and improves defense tactics. The platform utilizes adaptive learning mechanisms to adapt to emerging new patterns of attacks, which makes it capable of detecting even novel adversarial behavior.

Ultimately, Threat'Sentry hopes to close the gap between model performance and model security. With its trio of detection, simulation, and defense, it gives a complete solution to protecting ML models from manipulation and data poisoning. With the inclusion of explainability, robustness assessment, and automated countermeasure deployment, Threat'Sentry promotes trust in AI-based systems and creates a proactive defense framework against adversarial attacks in contemporary machine learning pipeline

Chapter 1

Introduction

As machine learning (ML) systems power critical applications like autonomous driving and cybersecurity, their vulnerability to adversarial attacks poses serious risks. Threat'Sentry is an adaptive defense framework that detects, simulates, and protects ML models from such manipulations. By monitoring inputs, generating adversarial examples, and adapting to emerging threats, it strengthens model robustness and reliability. With built-in explainability and automated countermeasures, Threat'Sentry provides a comprehensive and proactive defense for securing modern AI systems.

1.1 Motivation

With the rapid proliferation of artificial intelligence (AI) and machine learning (ML) across domains such as healthcare, finance, autonomous systems, and cybersecurity, the reliability of these models has become paramount. However, ML models—particularly deep neural networks—are known to be vulnerable to adversarial attacks, where imperceptible perturbations to input data can lead to erroneous or catastrophic predictions. Such attacks not only compromise model accuracy but also threaten the integrity and trustworthiness of AI-driven systems. This vulnerability underscores the need for a comprehensive, automated, and adaptive framework that can continuously evaluate, detect, and defend against adversarial manipulations. Threat'Sentry is motivated by this necessity—to ensure that AI systems are not just intelligent, but resilient and secure against evolving adversarial threats.

1.2 Problem Analysis

- Traditional ML models are designed with performance optimization in mind but lack intrinsic mechanisms to recognize or mitigate adversarially perturbed inputs. Existing defense strategies—such as adversarial training, gradient masking, or input preprocessing—tend to be attack-specific and often degrade performance on clean data. Furthermore, many organizations .

deploy ML models without continuous security evaluation, leaving them exposed to unseen adversarial patterns and transfer-based black-box attacks. The absence of a unified framework that can detect adversarial anomalies, simulate diverse attack types, and dynamically adapt defense strategies presents a significant research and practical challenge.

1.3 Objectives

- To identify adversarial perturbations in model inputs through anomaly analysis of activations, gradients, and feature distribution.
- To generate and replicate a wide range of adversarial attacks—white-box, gray-box, and black-box—for robustness benchmarking.
- To deploy adaptive countermeasures such as adversarial training, feature squeezing, and denoising to enhance model resilience.
- To develop standardized robustness metrics and visualizations that quantify the model's vulnerability under various threat models.
- To provide a modular, scalable architecture that seamlessly integrates with existing ML pipelines for continuous security monitoring.

1.4 Scope

The scope of Threat'Sentry encompasses the comprehensive study and implementation of mechanisms to detect, simulate, and defend against adversarial attacks targeting machine learning models. The framework is designed to operate across various ML paradigms, including supervised and deep learning models used in image recognition, natural language processing, and structured data classification. It provides a modular architecture for integrating adversarial attack generators, anomaly detectors, and adaptive defense strategies. Additionally, Threat'Sentry enables continuous robustness evaluation through visualization dashboards and performance metrics, allowing researchers and practitioners to assess model vulnerability and defense efficacy in real-time. The project aims to advance adversarial robustness research while maintaining practical deployability within the existing AI pipeline.

Chapter 2

Literature Review

2.1 Related Work

[1] Adversarial Deep Learning: A Survey on Adversarial Attacks and Defense Mechanisms on Image Classification (2022)

Summary: This paper surveys adversarial attacks and defense mechanisms specifically in image classification. It covers common attack strategies such as FGSM, PGD, and C&W, along with defensive methods including adversarial training, input preprocessing, and detection techniques. The study emphasizes evaluation metrics to assess model robustness and highlights challenges in defending against adaptive attacks.

Relevance to Threat'Sentry:

- **Threat Simulation:** Provides a comprehensive overview of attack techniques, useful for designing Threat'Sentry's simulation engine.

- **Defense Benchmarking:** Offers structured evaluation metrics, informing Threat'Sentry's robustness assessment and verification modules.

[2] Machine Learning Adversarial Attacks: A Survey Beyond (2022)

Summary: This survey extends beyond image-based attacks to include text, tabular data, and networked ML systems. It classifies attacks as white-box, black-box, targeted, and untargeted, and reviews defense mechanisms across different domains. The paper highlights the need for standardized evaluation frameworks to benchmark model robustness across data types.

Relevance to Threat'Sentry:

- Cross-Domain Simulation: Supports Threat'Sentry in simulating attacks across multiple data modalities.
- Defense Benchmarking: Provides guidance on evaluation metrics for defenses across diverse data types.

[3] A Systematic Review of Adversarial Machine Learning Attacks, Defensive Controls, and Technologies (2024)

Summary: This paper provides a lifecycle-oriented review of adversarial ML, covering attacks, defenses, and security gaps from training to deployment. It details threat detection methods, model hardening strategies, and runtime monitoring techniques, while emphasizing gaps in current defenses against adaptive and novel attacks.

Relevance to Threat'Sentry:

- **Lifecycle-Based Defense:** Helps design Threat'Sentry's continuous monitoring and adaptive defense mechanisms.
- **Threat Simulation:** Guides modeling of attacks at multiple stages of the ML lifecycle to proactively identify vulnerabilities.

[4] Adversarial Attacks and Defenses in Machine Learning-Empowered Communication Systems and Networks: A Contemporary Survey (2023)

Summary: This paper surveys attacks and defenses in ML for communication systems and networks, including evasion, poisoning, and model inversion attacks. Defensive strategies such as adversarial training, detection algorithms, and secure model design are discussed, along with metrics for robustness evaluation in networked ML systems.

Relevance to Threat'Sentry:

- **Domain-Specific Simulation:** Useful for Threat'Sentry if deployed in communication systems or cyber-physical networks.
- **Defense Benchmarking:** Provides evaluation frameworks for measuring robustness in networked ML applications.

[5] Evaluating and Improving Adversarial Robustness of Machine Learning-Based Network Intrusion Detectors (2021)

Summary: This paper focuses on the adversarial robustness of ML models used for network intrusion detection. It examines attack strategies targeting classifiers and evaluates defense mechanisms such as adversarial training, preprocessing, and ensemble methods. Metrics for quantifying robustness and resilience of intrusion detection systems are proposed.

Relevance to Threat'Sentry:

- **Threat Simulation:** Supplies real-world network attack scenarios for Threat'Sentry's simulation engine.
- **Defense Benchmarking:** Provides methods to evaluate and improve ML-based defenses, directly informing Threat'Sentry's robustness modules.

2.2 Existing System

In the rapidly advancing domain of machine learning security and model reliability, several frameworks and systems have emerged that align closely with the objectives of the Threat'Sentry project. These solutions, ranging from early-stage adversarial attack libraries to advanced AI security and robustness evaluation platforms, provide valuable insights into the current state of adversarial defense research. By examining these existing systems, Threat'Sentry seeks to bridge the gap between academic adversarial research and practical, deployable defense mechanisms, offering a unified platform for detection, simulation, and adaptive mitigation of adversarial threats across diverse ML models.

1. Traditional Adversarial Defense Frameworks

Traditional frameworks like Cleverhans and Foolbox primarily focus on implementing adversarial attack algorithms for research and benchmarking. While they offer valuable insights into model vulnerabilities, they lack integrated, real-time defense mechanisms or continuous monitoring capabilities. These tools are primarily research-focused and do not extend to adaptive defense or operational deployment, limiting their practicality in production-grade ML systems.

2. Modern Adversarial Defense Platforms

Recent platforms such as Adversarial Robustness Toolbox (ART) and Adversarial ML Threat Matrix (Microsoft) have introduced systematic ways to evaluate and mitigate adversarial threats. They support multiple attack methods, training defenses, and standardized robustness reporting. However, these systems still operate as static frameworks rather than dynamic platforms that evolve with adversarial trends, leaving a gap in continuous defense automation.

3. AI Security Testing and Robustness Evaluation Systems

AI security testing solutions like RobustBench and RobustML provide benchmarking datasets and leaderboards for adversarial robustness evaluation. While they advance reproducibility and comparative analysis, they lack modular architectures for attack simulation or live model protection. Their primary focus remains on empirical research rather than integrated deployment environments.

4. Automated Adversarial Detection Systems

Emerging research prototypes have begun integrating adversarial detection into production environments, using anomaly detection and behavior-based triggers. However, many of these rely heavily on supervised detection models and struggle with generalizing to novel attack types. Threat'Sentry differentiates itself by combining dynamic simulation, adaptive defense, and hybrid detection within a single, scalable platform capable of continuous learning and self-improvement.

5. Hybrid Security-Oriented AI Platforms

Recent advancements in AI security, such as ShieldAI and AI Firewall frameworks, attempt to bridge ML monitoring with cybersecurity principles. These systems employ policy-driven controls to block adversarial traffic but often overlook model-specific perturbations. Threat'Sentry extends this concept by embedding security intelligence directly into the ML model lifecycle—covering training, evaluation, and deployment phases

Chapter 3

Proposed System

Threat'Sentry provides the following core functionalities that make adversarial threat analysis and defense a seamless, automated, and adaptive process for machine learning systems:

1. Adversarial Attack Simulation

Threat'Sentry includes a powerful simulation engine capable of generating adversarial examples using multiple attack strategies such as FGSM, PGD, DeepFool, and CW attacks. This enables developers and researchers to evaluate the robustness of their models against both white-box and black-box adversarial threats, helping identify vulnerabilities before deployment.

2. Adaptive Defense Mechanism

The system integrates dynamic defense techniques that adapt to evolving adversarial patterns through adversarial training, feature denoising, and gradient regularization. By continuously learning from simulated attack data, Threat'Sentry enhances model resilience over time, reducing susceptibility to new or unseen adversarial inputs.

3. Real-Time Threat Detection and Monitoring

Threat'Sentry features a real-time detection module that monitors model behavior and input patterns to detect anomalies indicative of adversarial interference. Techniques such as gradient-based analysis, feature squeezing, and activation anomaly detection are employed to identify adversarial attacks as they occur during inference, ensuring prompt defensive response.

4. Explainable and Visual Analysis

To promote transparency and interpretability, Threat'Sentry integrates visualization tools like saliency mapping and gradient interpretation (e.g., Grad-CAM, SHAP). These visual

explanations help users understand how and why a model's predictions are being influenced by potential adversarial perturbations, supporting trust and accountability in AI decisions.

5. Automated Defense Evaluation and Reporting

The system provides a comprehensive evaluation framework that benchmarks model robustness across various attack intensities and datasets. It automatically generates detailed security reports highlighting model weaknesses, achieving robustness scores, and recommended defense improvements, streamlining the audit and compliance process.

3.1 Proposed Approach and Details

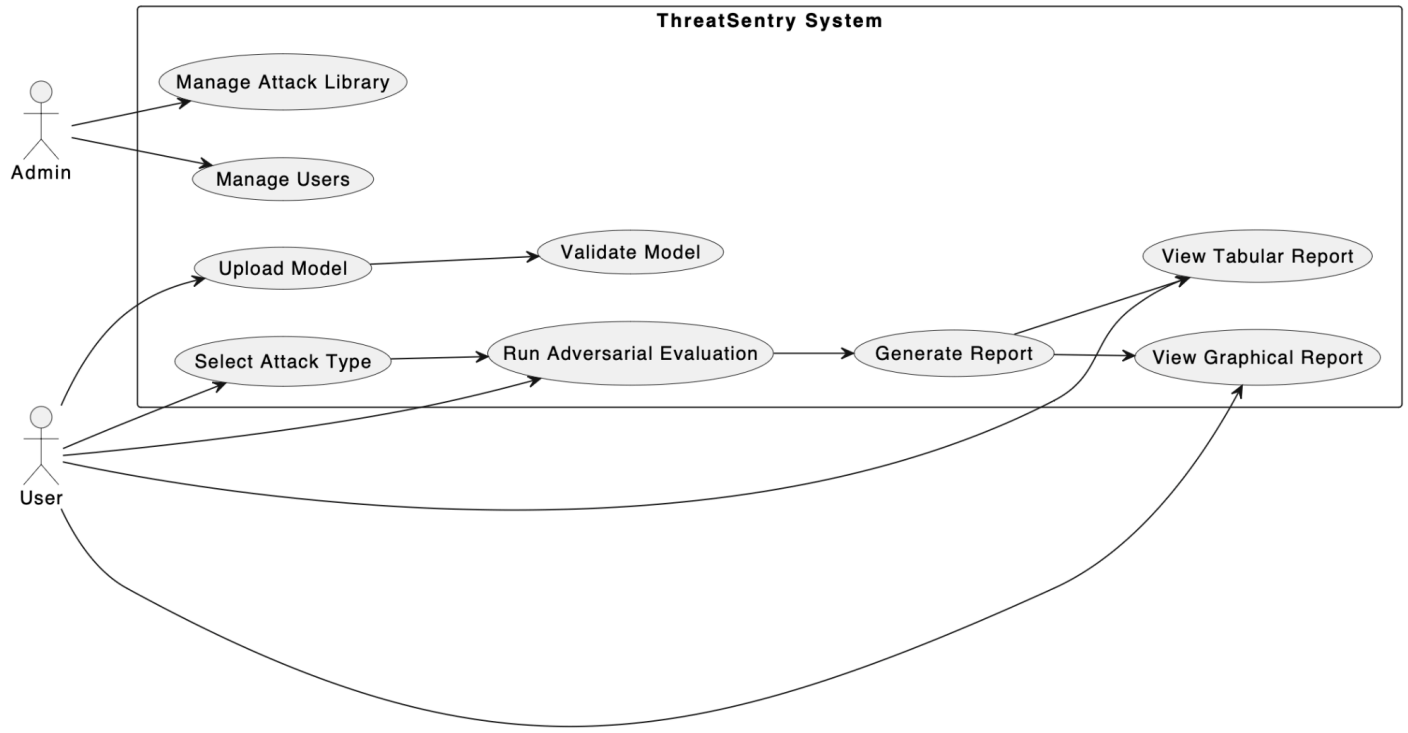

Fig 3.1.1 Use-Case Diagram

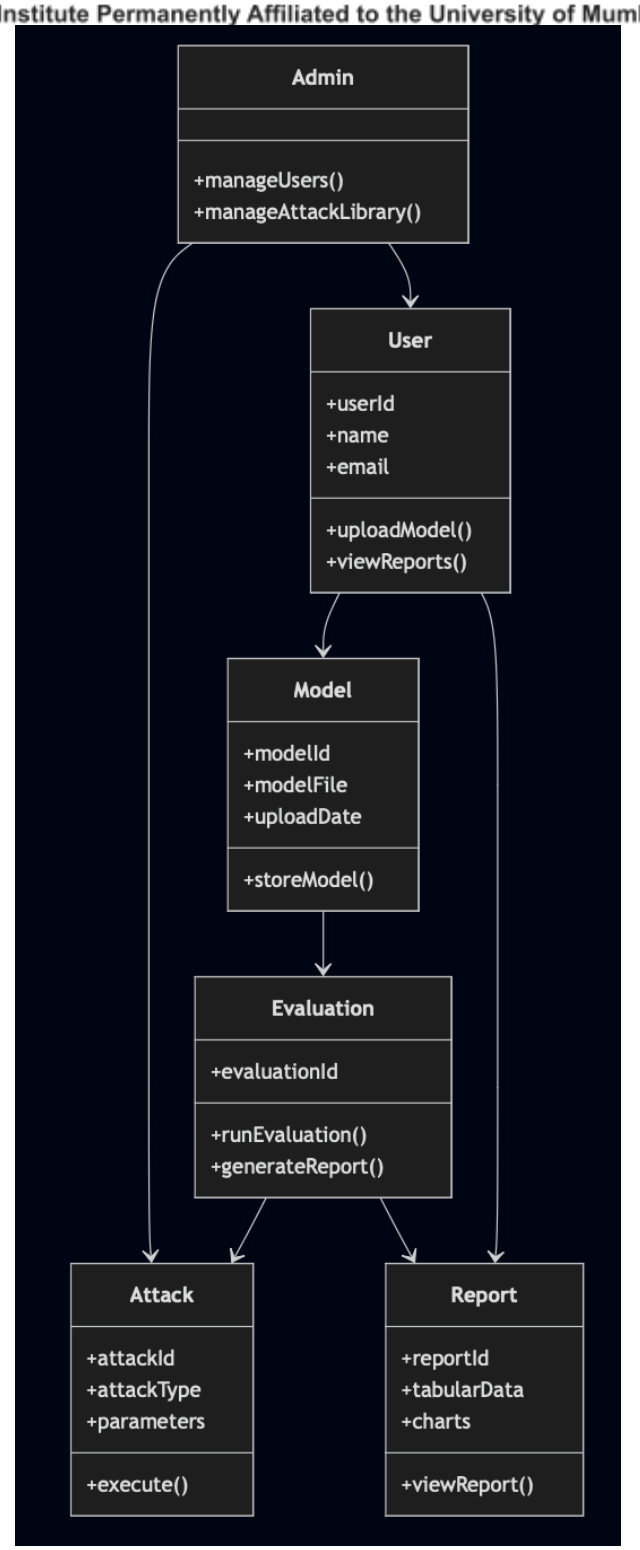

Fig 3.1.2 Class Diagram

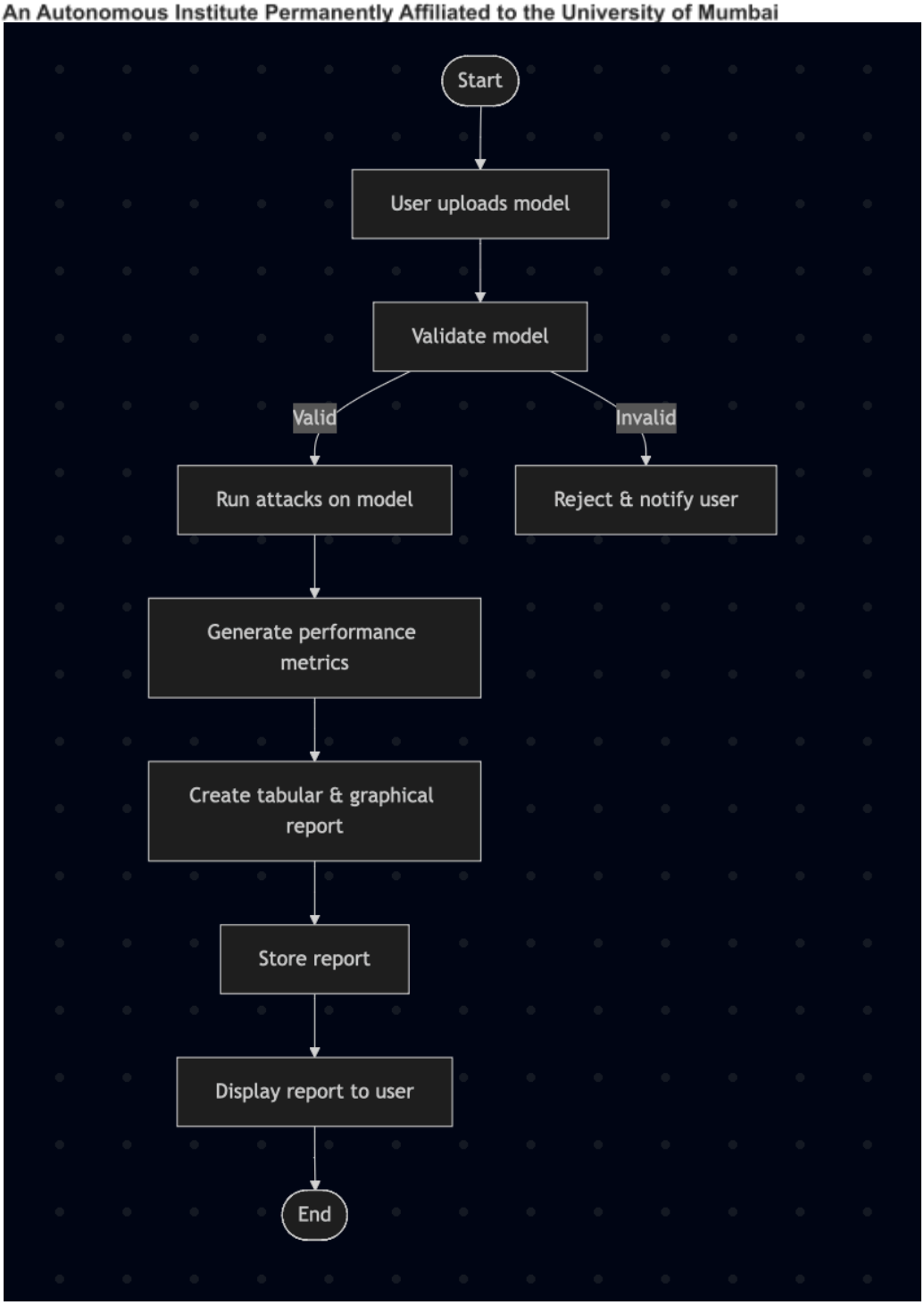

Fig.3.1.3 Activity Diagram

3.2 Innovation in Idea

The Threat'Sentry project introduces several innovative aspects that distinguish it from conventional ML monitoring and security systems:

- **Real-Time Intelligent Threat Detection:** Unlike standard ML monitoring tools that only track model performance metrics, Threat'Sentry detects adversarial attacks on ML models in real-time. It identifies anomalies in input data, model predictions, and feature patterns, and routes alerts to the most relevant ML engineer or security analyst for immediate action.
- **Multi-Channel Attack Alerts with Contextual Priority:** Threat'Sentry integrates email, SMS, and dashboard notifications, intelligently prioritizing alerts based on the type and severity of the adversarial attack. This ensures that high-risk attacks, such as data poisoning or evasion attacks, are addressed immediately while reducing unnecessary alert noise.
- **Hybrid Monitoring Approach:** The system combines statistical monitoring of model outputs with adversarial robustness testing. This hybrid approach enables detection of both subtle input manipulations and large-scale attacks, providing comprehensive protection beyond conventional model evaluation metrics.
- **Intelligent Threat Aggregation:** Threat'Sentry clusters related adversarial events to prevent multiple redundant alerts for similar attack vectors. This provides a clear, consolidated view of attack patterns, improving the efficiency of mitigation strategies.
- **Scalable Modular Architecture:** The platform is built with modular ML monitoring and alerting agents that can scale horizontally, supporting multiple models, datasets, and deployment environments without performance degradation.
- **Proactive Insights and Historical Analytics:** Threat'Sentry logs all adversarial events and model responses, generating reports on attack trends, model vulnerabilities, and robustness metrics. This allows teams to proactively strengthen model defenses and improve reliability against future attacks.

3.3 Timeline

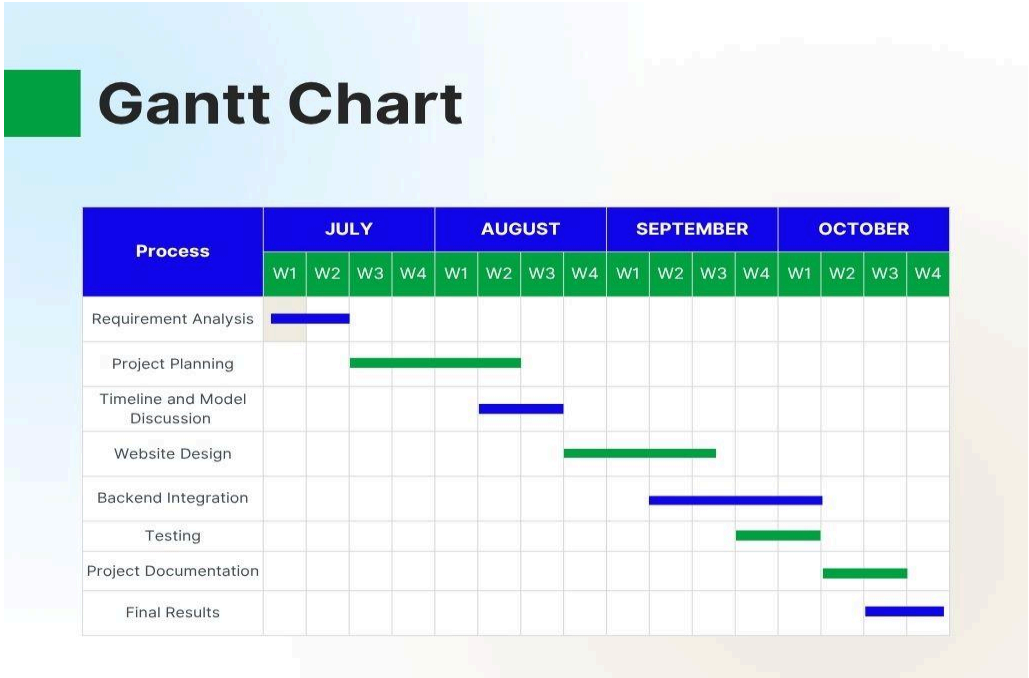

Fig 3.3.1 - Gantt Chart

3.4 Roles and Responsibilities

Neelaksha Bhardwaj: Creating the dataset, training model and backend of the project.

Kshitij Bramhecha: Creating the Frontend and Authentication and working on scripting of ML models of the project.

Jakumar Gor: Planning the project, Creating train and testing data for model, and

Chapter 4

Implementation Details and Results

4.1 Technology Stack

Frontend - The frontend of Threat'Sentry is built using ReactJS, providing a dynamic and responsive user interface for monitoring adversarial attacks on ML models. To enhance the UI/UX, the system leverages ShadCN components, which allow for modular, reusable, and visually appealing interface elements, ensuring that security analysts can easily track alerts, view consolidated incidents, and access detailed analytics in real-time.

Backend - On the backend, the platform is powered by Python, which handles the core logic for real-time threat detection, alert prioritization, and intelligent aggregation of adversarial events. Docker is used to containerize the backend services and ML monitoring agents, ensuring consistency across environments, easy deployment, and horizontal scalability to handle multiple models and datasets efficiently. The system also integrates AI/ML modules for anomaly detection, adversarial input analysis, and predictive insights, enabling Threat'Sentry to not only detect attacks but also provide actionable recommendations for strengthening model robustness.

Overall, this tech stack combines a modern, responsive frontend with a robust, scalable backend capable of real-time AI-driven threat detection, making Threat'Sentry a comprehensive solution for securing machine learning models against adversarial threats.

4.2 Implementation Parameters

The implementation of Threat'Sentry involves configuring multiple parameters to ensure accurate detection and efficient handling of adversarial attacks on ML models. Monitoring frequency is a key parameter, defining how often the system evaluates incoming inputs and model outputs for anomalies. Alert thresholds are critical for determining the sensitivity of detection modules, balancing between catching subtle adversarial manipulations and minimizing false positives to ensure actionable alerts.

Algorithmic parameters play a central role in Threat'Sentry's detection accuracy. These include feature weighting, which assigns importance to different input features, and anomaly scoring thresholds, which define the cutoff for considering a deviation as a potential attack. Other parameters include distance metrics used in detecting out-of-distribution inputs, gradient-based attack detection sensitivity, and confidence score thresholds from the ML model's predictions. Fine-tuning these parameters allows the system to detect both known and zero-day adversarial attacks effectively.

4.3 Preliminary Results

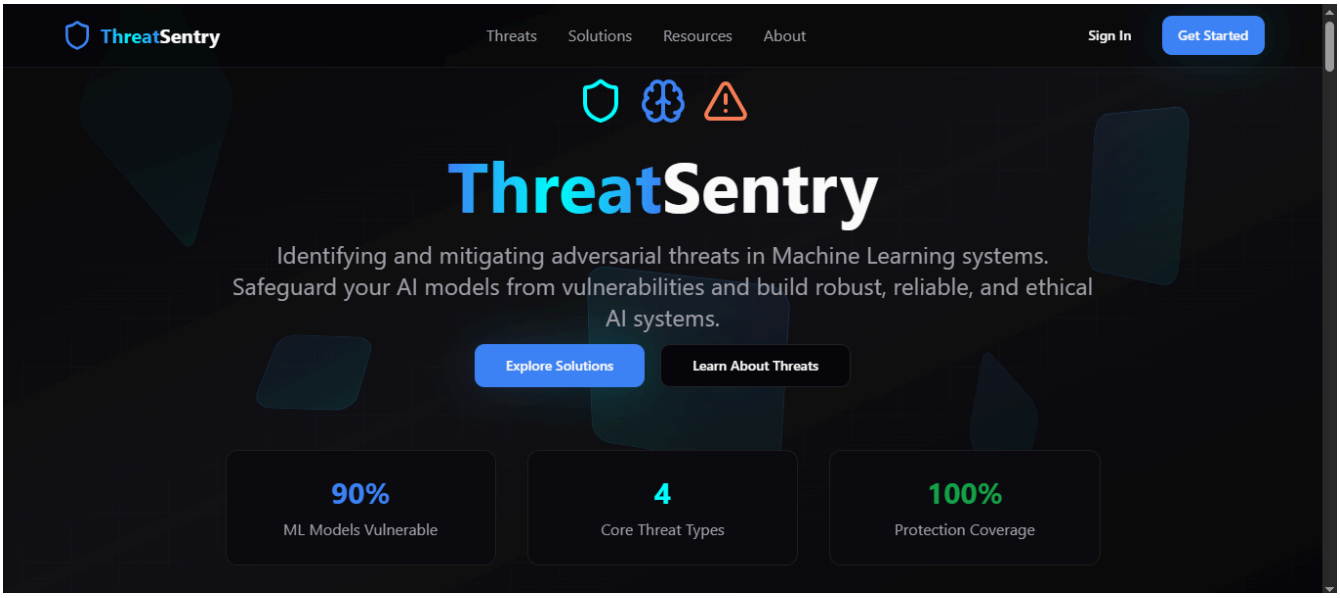

Fig 4.3.1 - Introduction Page to ThreatSentry.

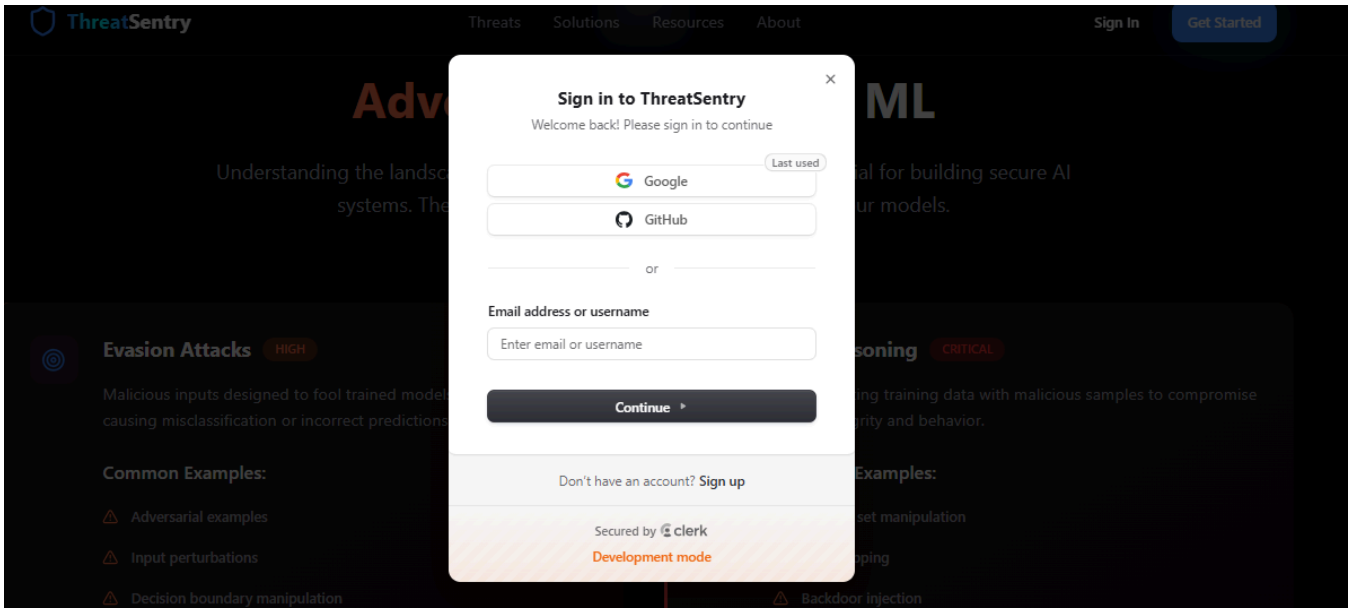

Fig 4.3.2 - SignUp/Login page for authentication.

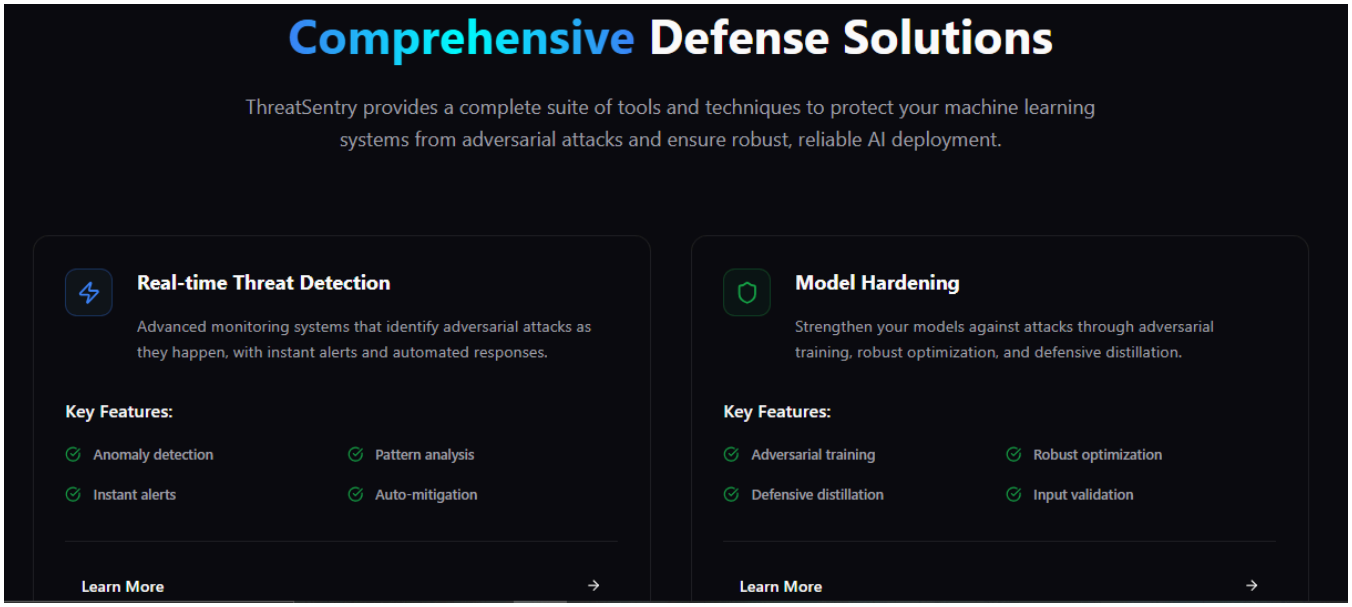

Fig 4.3.3 - Features provided by ThreatSentry.

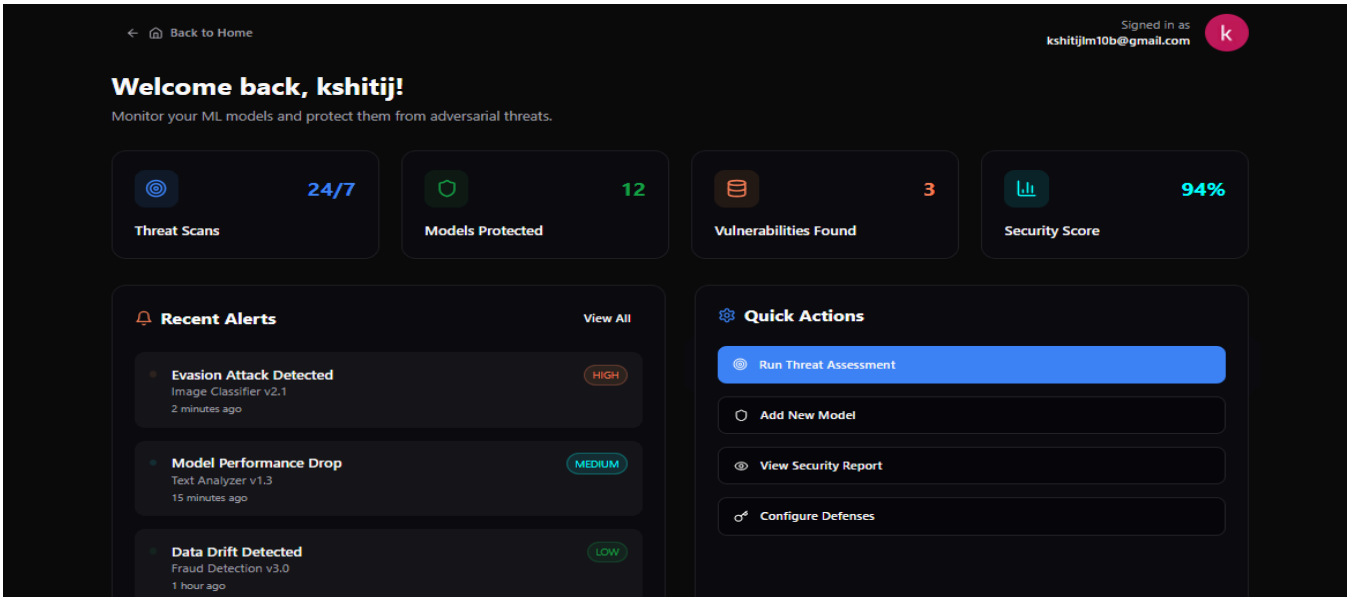

Fig 4.3.4 - Dashboard Page to show recent threat assessments for the ml models.

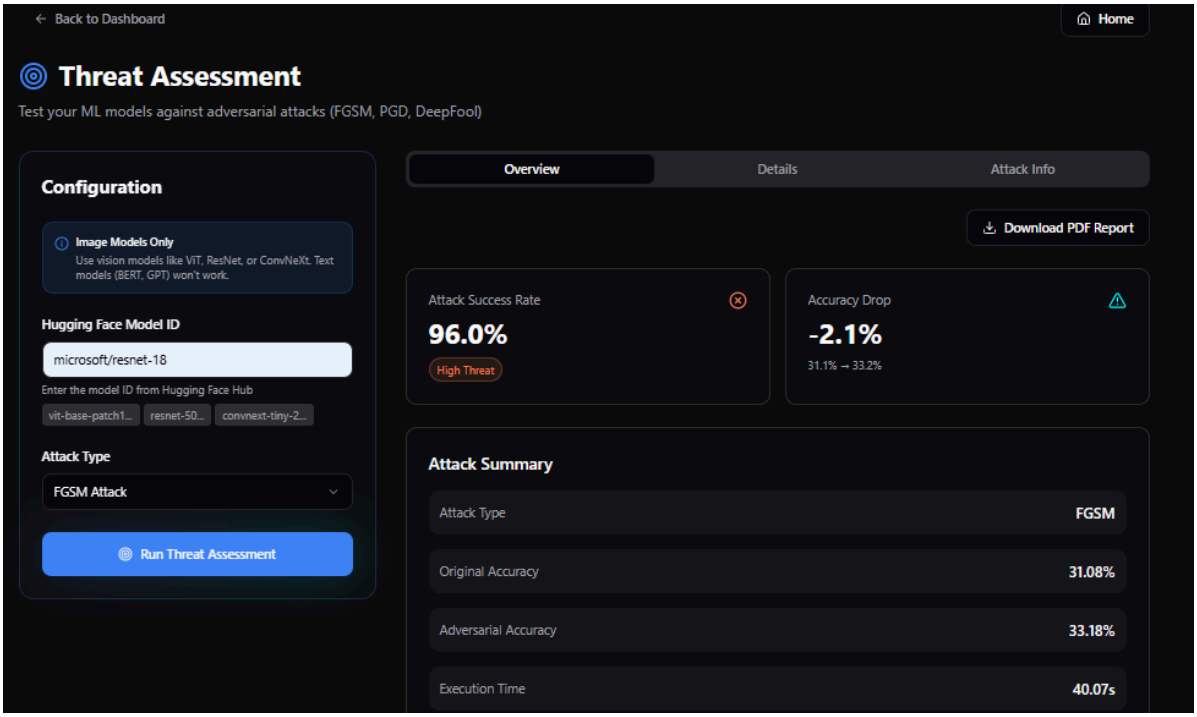

Fig 4.3.5 - Testing ML Model for Threat Assessment.

Chapter 5: Conclusion

5.1 Conclusion

Threat'Sentry represents a significant step forward in securing machine learning models against adversarial threats. By combining real-time detection, hybrid monitoring, and intelligent alert aggregation, it goes beyond traditional monitoring systems to actively identify and respond to subtle attacks that could compromise model integrity. The platform's scalable and modular design ensures that it can efficiently handle multiple models, datasets, and deployment environments, making it suitable for a wide range of applications—from startups to large enterprises.The system's proactive insights and historical analytics enable teams to understand attack trends and model vulnerabilities, allowing for informed decisions to enhance robustness and reliability. By prioritizing alerts based on severity and context, Threat'Sentry reduces response times and ensures accountability among ML engineers and security analysts. Overall, it provides a comprehensive and intelligent solution to one of the most pressing challenges in modern AI deployment—maintaining trust and performance under adversarial conditions. Its ability to consolidate related adversarial events and reduce alert fatigue further streamlines security operations, demonstrating a thoughtful approach to both technical effectiveness and user experience.

5.2 Future Plan

In the future, Threat'Sentry can be extended to incorporate automated mitigation strategies, where detected adversarial inputs are immediately neutralized or quarantined before affecting model predictions. Integrating reinforcement learning and adaptive defense mechanisms could enable models to learn from attacks in real-time, improving robustness without manual intervention.Additionally, expanding the system to support cross-model threat intelligence sharing would allow multiple models and deployments to benefit from detected attack patterns globally. Coupled with AI-driven predictive analytics, Threat'Sentry could evolve into a proactive security platform capable of anticipating adversarial strategies and dynamically hardening ML models against emerging threats.

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